



# OPERATING SYSTEMS

## UE24CS242B

### Multithreading Models, Thread Creation and Scheduling

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**Pavan A C**

Department of Computer Science & Engineering

# OPERATING SYSTEMS

## Slides Credits for all the PPTs of this course

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- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:
  1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9<sup>th</sup> edition 2013 and some slides from 10<sup>th</sup> edition 2018
  2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9<sup>th</sup> edition 2018
  3. Some presentation transcripts from A. Frank – P. Weisberg
  4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau



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## Multithreading Models, Thread Creation and Scheduling

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## Multithreading Models

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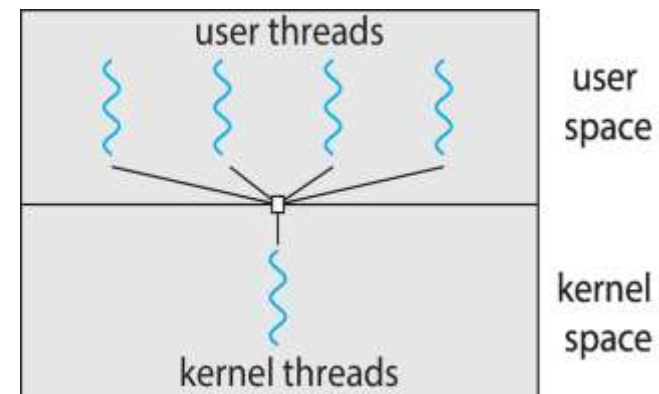
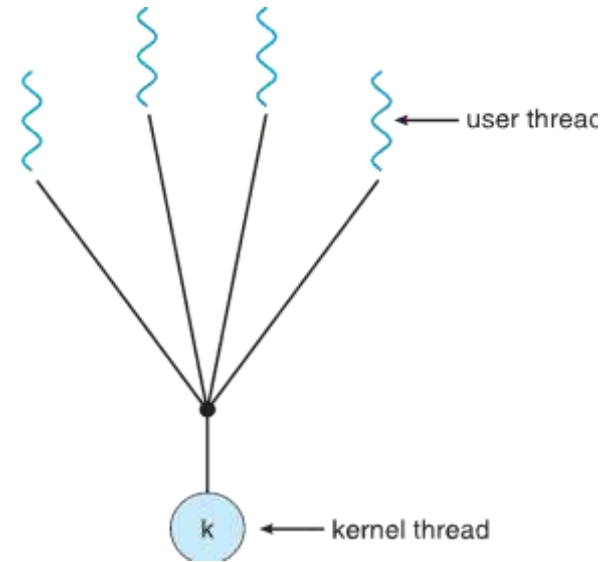
- Many-to-One
- One-to-One
- Many-to-Many



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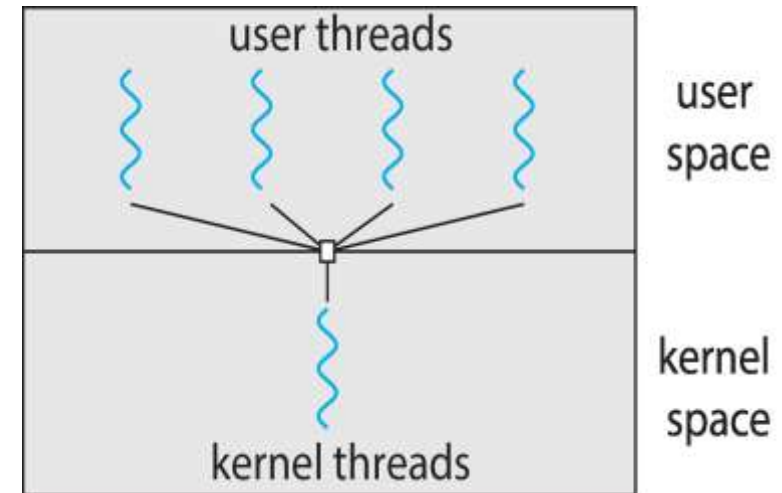
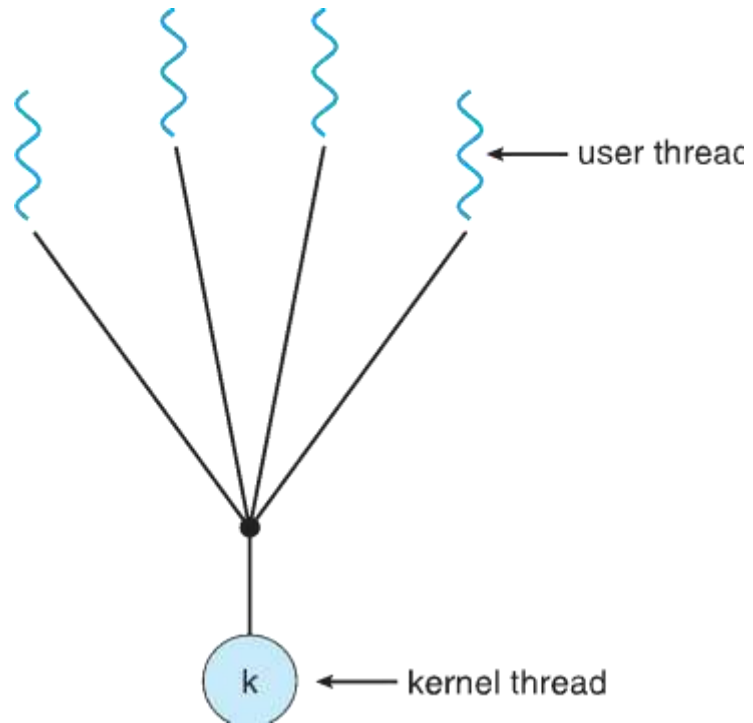
## Many-to-One

- Many user-level threads mapped to single kernel thread
- One thread blocking causes all to block
- Multiple threads may not run in parallel on multicore system because only one may be in kernel at a time
- Few systems currently use this model
- Examples:
  - **Solaris Green Threads**
  - **GNU Portable Threads**



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## Many-to-One



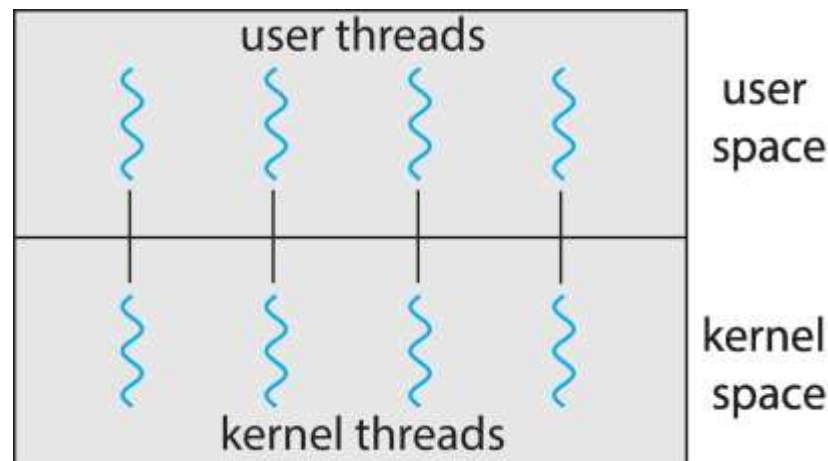
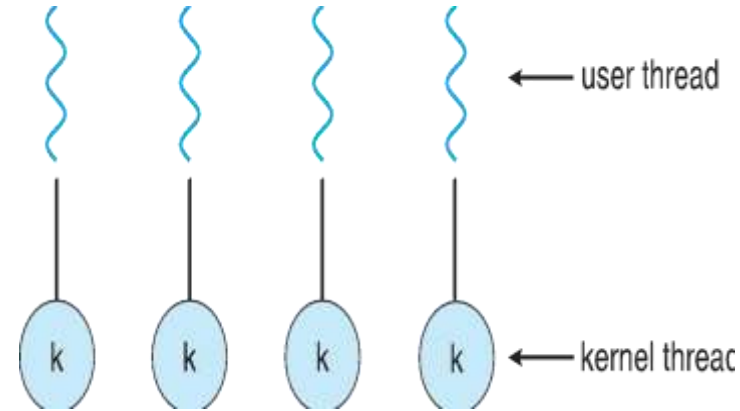
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## One-to-One

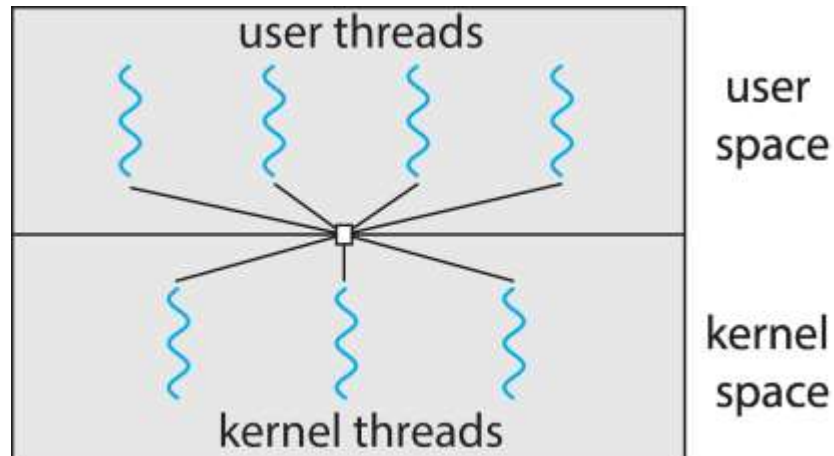
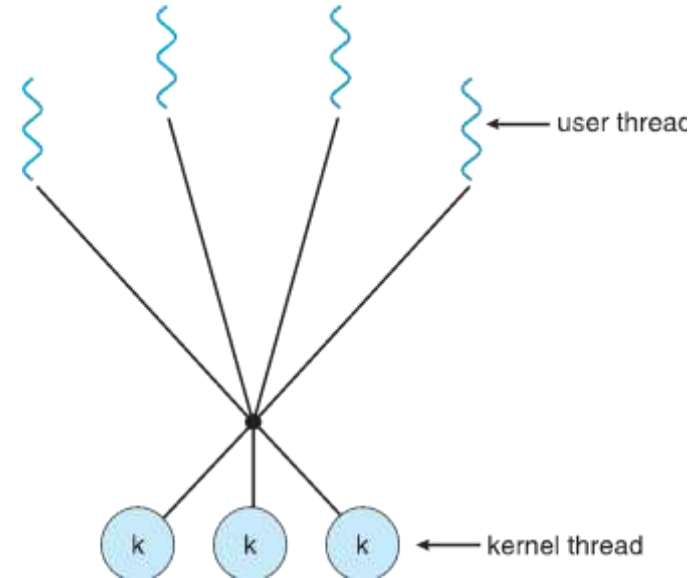
- Each user-level thread maps to kernel thread
- Creating a user-level thread creates a kernel thread
- More concurrency than many-to-one
- Number of threads per process sometimes restricted due to overhead
- Examples
  - Windows
  - Linux
  - Solaris 9 and later



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## Many-to-Many Model

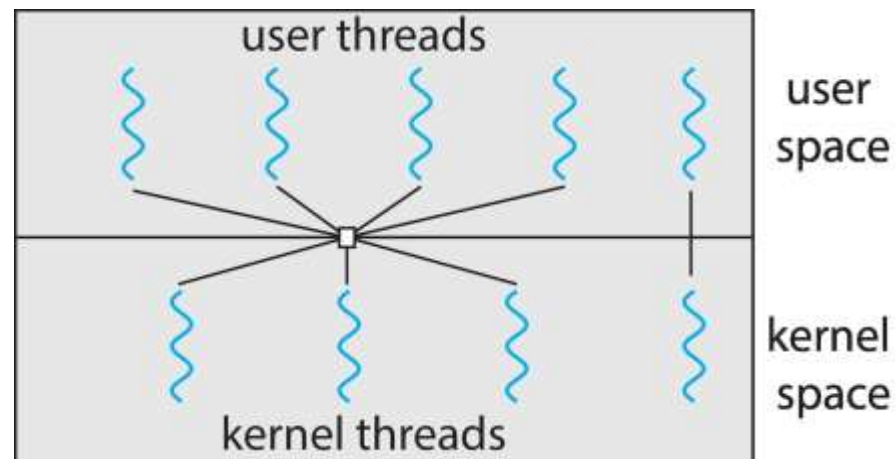
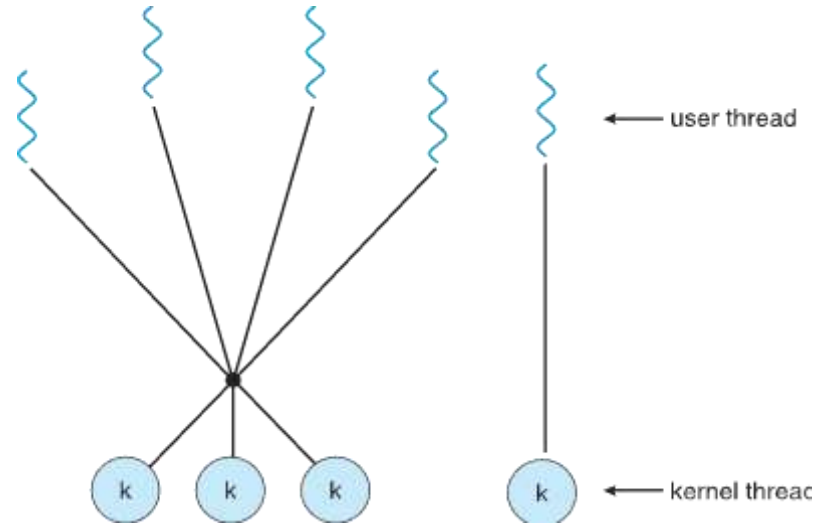
- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Solaris prior to version 9
- Windows with the *ThreadFiber* package



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## Two-level Model

- Similar to M:M, except that it allows a user thread to be **bound** to kernel thread
- Examples
  - IRIX
  - HP-UX
  - Tru64 UNIX
  - Solaris 8 and earlier



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## Pthreads

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- May be provided either as user-level or kernel-level
- A POSIX standard (IEEE 1003.1c) API for thread creation and synchronization
- ***Specification***, not ***implementation***
- API specifies behavior of the thread library, implementation is up to development of the library
- Common in UNIX operating systems (Solaris, Linux, Mac OS X)



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## Pthreads Example

```
#include <pthread.h>
#include <stdio.h>

int sum; /* this data is shared by the thread(s) */
void *runner(void *param); /* threads call this function */

int main(int argc, char *argv[])
{
    pthread_t tid; /* the thread identifier */
    pthread_attr_t attr; /* set of thread attributes */

    if (argc != 2) {
        fprintf(stderr, "usage: a.out <integer value>\n");
        return -1;
    }
    if (atoi(argv[1]) < 0) {
        fprintf(stderr, "%d must be >= 0\n", atoi(argv[1]));
        return -1;
    }
}
```



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## Pthreads Example

```
/* get the default attributes */
pthread_attr_init(&attr);
/* create the thread */
pthread_create(&tid,&attr,runner,argv[1]);
/* wait for the thread to exit */
pthread_join(tid,NULL);

printf("sum = %d\n",sum);
}

/* The thread will begin control in this function */
void *runner(void *param)
{
    int i, upper = atoi(param);
    sum = 0;

    for (i = 1; i <= upper; i++)
        sum += i;

    pthread_exit(0);
}
```



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## Pthreads Code for Joining 10 Threads

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```
#define NUM_THREADS 10

/* an array of threads to be joined upon */
pthread_t workers[NUM_THREADS];

for (int i = 0; i < NUM_THREADS; i++)
    pthread_join(workers[i], NULL);
```



- Distinction between user-level and kernel-level threads
- When threads supported, threads scheduled, not processes
- Many-to-one and many-to-many models, thread library schedules user-level threads to run on LWP.
- Scheduling of user level threads (ULT) to kernel level threads (KLT) via leight-weight process (LWP) by the application developer.
  - **Leightweight Process (LWP)** : Light-weight process are threads in the user space that acts as an interface for the ULT to access the physical CPU resources
- Scheduling of kernel level threads by the system scheduler to perform different unique OS functions.

- Many-to-one and many-to-many models, thread library schedules user-level threads to run on LWP
  - Known as **process-contention scope (PCS)** since scheduling competition is within the process
  - Typically done via priority set by programmer
- Kernel thread scheduled onto available CPU is **system-contention scope (SCS)**
  - competition among all threads in system

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## Pthread Scheduling

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- API allows specifying either PCS or SCS during thread creation
  - PTHREAD\_SCOPE\_PROCESS schedules threads using PCS scheduling
  - PTHREAD\_SCOPE\_SYSTEM schedules threads using SCS scheduling
- Can be limited by OS – Linux and Mac OS X only allow PTHREAD\_SCOPE\_SYSTEM



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## Pthread Scheduling API

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```
#include <pthread.h>
#include <stdio.h>
#define NUM_THREADS 5
int main(int argc, char *argv[]) {
    int i, scope;
    pthread_t tid[NUM_THREADS];
    pthread_attr_t attr;
    /* get the default attributes */
    pthread_attr_init(&attr);
    /* first inquire on the current scope */
    if (pthread_attr_getscope(&attr, &scope) != 0)
        fprintf(stderr, "Unable to get scheduling scope\n");
    else {
        if (scope == PTHREAD_SCOPE_PROCESS)
            printf("PTHREAD_SCOPE_PROCESS");
        else if (scope == PTHREAD_SCOPE_SYSTEM)
            printf("PTHREAD_SCOPE_SYSTEM");
        else
            fprintf(stderr, "Illegal scope value.\n");
    }
}
```



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## Pthread Scheduling API

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```
/* set the scheduling algorithm to PCS or SCS */
pthread_attr_setscope(&attr, PTHREAD_SCOPE_SYSTEM);
/* create the threads */
for (i = 0; i < NUM_THREADS; i++)
    pthread_create(&tid[i], &attr, runner, NULL);
/* now join on each thread */
for (i = 0; i < NUM_THREADS; i++)
    pthread_join(tid[i], NULL);
}
/* Each thread will begin control in this function */
void *runner(void *param)
{
    /* do some work ... */
    pthread_exit(0);
}
```





# THANK YOU

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**Pavan A C**

Department of Computer Science & Engineering

**pavanac@pes.edu**